

|                                                                   |
|-------------------------------------------------------------------|
| <b>Wk0500 -- ET-ESP32-RS485 V2 Board + Arduino Cloud Platform</b> |
| <b>Std01.รหัส B6729615 ชื่อ-สกุล ปกักร สมสุข</b>                  |
| <b>Std01.รหัส B6728960 ชื่อ-สกุล มงคล น้อยบาท</b>                 |

|                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                             |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1. Read &gt;&gt; XY-MD02</b> <ul style="list-style-type: none"> <li>- Hardware - A</li> <li>- Show with <b>Modbus Poll (Serial)</b></li> <li>- 9600, N, 8, 1</li> <li>- Func=04, Start=1, Quntity=2</li> </ul>               | <b>2. Read &gt;&gt; XY-MD02</b> <ul style="list-style-type: none"> <li>- Hardware - B, ESP32 Board</li> <li>- Show with <b>Blynk2.0</b></li> </ul>                                                                                                          |
| <b>3. Read &gt;&gt; REG48PUN2LJHU</b> <ul style="list-style-type: none"> <li>- Hardware - XS</li> <li>- Show with <b>Modbus Poll (Serial)</b></li> </ul>                                                                        | <b>4. Read &gt;&gt; REG48PUN2LJHU</b> <ul style="list-style-type: none"> <li>- Hardware - XU</li> <li>- Show with PV, SV on <b>HMI (SK-070HS) via RS485</b></li> </ul>                                                                                      |
| <b>5. Control and Monitor</b> <ul style="list-style-type: none"> <li>- Hardware - C</li> <li>- Control Monitor ข้อ 10</li> <li>- Control and monitor by <b>Arduino IoT Platform</b></li> </ul>                                  | <b>6. Control and Monitor</b> <ul style="list-style-type: none"> <li>- Hardware - XT</li> <li>- Control Monitor ข้อ 10</li> <li>- Display REG48PUN2LJHU</li> <li>- Display XY-MD02</li> <li>- Control and monitor by <b>Arduino IoT Platform</b></li> </ul> |
| <b>7. Modbus TCP</b> <ul style="list-style-type: none"> <li>- Hardware - XY</li> <li>- Show with <b>Modbus Poll (TCP)</b></li> <li>- Read REG48PUN2LJHU</li> <li>- Read XY-MD02</li> <li>- Control Monitor ตามข้อ 10</li> </ul> | <b>8. Modbus TCP with HMI</b> <ul style="list-style-type: none"> <li>- Hardware - XZ</li> <li>- Show with <b>HMI(SK-070HS) via Modbus TCP</b></li> <li>- Read REG48PUN2LJHU</li> <li>- Read XY-MD02</li> <li>- Control Monitor ตามข้อ 10</li> </ul>         |

Read:

- <https://www.h-schmidt.net/FloatConverter/IEEE754.html>
- <https://th.rs-online.com/web/p/temperature-controllers/8223416>
- SK-070 Serial Pinout = <https://www.samkoon.store/sk-hmi-structure/>

## Modbus TCP Client

## Modbus Poll

1. Read >> XY-MD02

- Hardware - A
- Show with **Modbus Poll (Serial)**
- 9600, N, 8, 1

Func=04, Start=1, Quntity=2

The screenshot displays the Modbus Poll - Mbpoll1 application window. The main interface shows a table with columns 'Name' and 'Value'. The 'Name' column contains '00000' and the 'Value' column contains '319'. Below this, a list of addresses from 0 to 9 is shown, with '319' highlighted in blue. The 'Communication Traffic' window is open, showing a list of transactions. The status bar at the bottom indicates 'Port 4: 9600-8-N-1'.

| Name  | Value |
|-------|-------|
| 00000 | 319   |
| 0     |       |
| 1     |       |
| 2     | 498   |
| 3     |       |
| 4     |       |
| 5     |       |
| 6     |       |
| 7     |       |
| 8     |       |
| 9     |       |

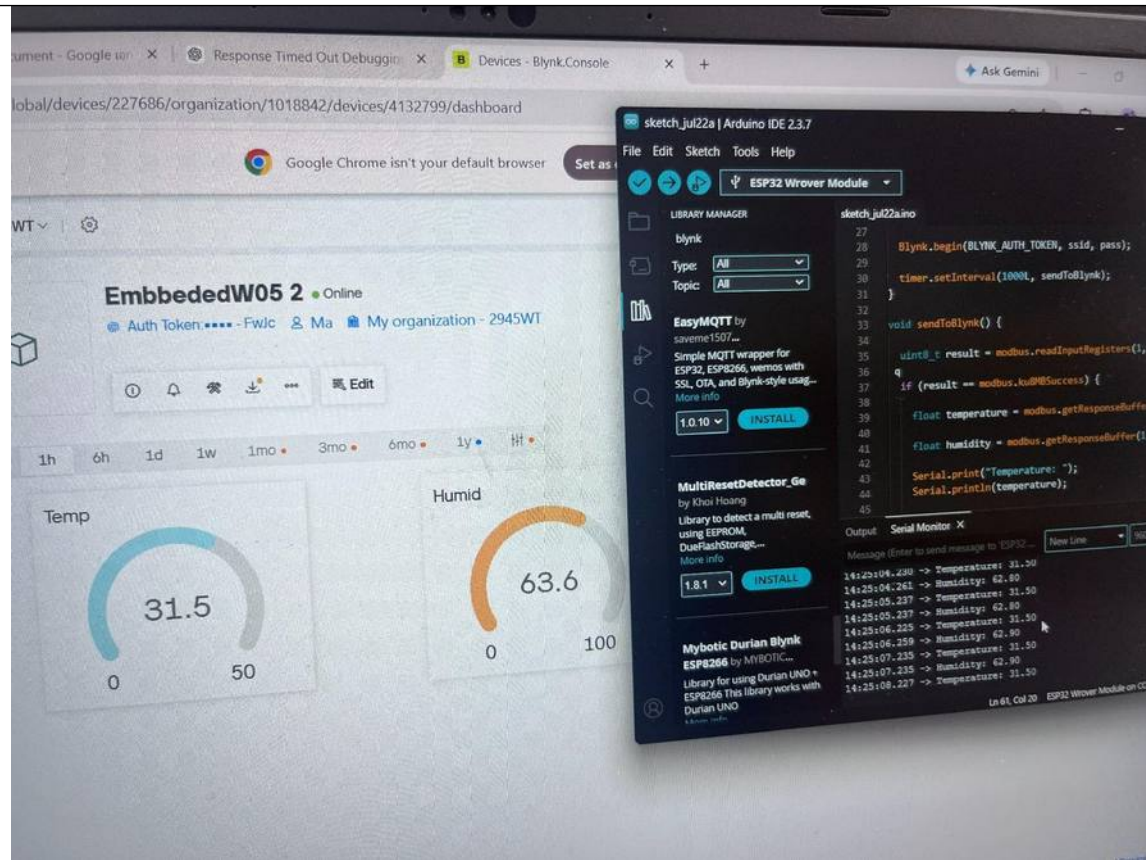
Communication Traffic

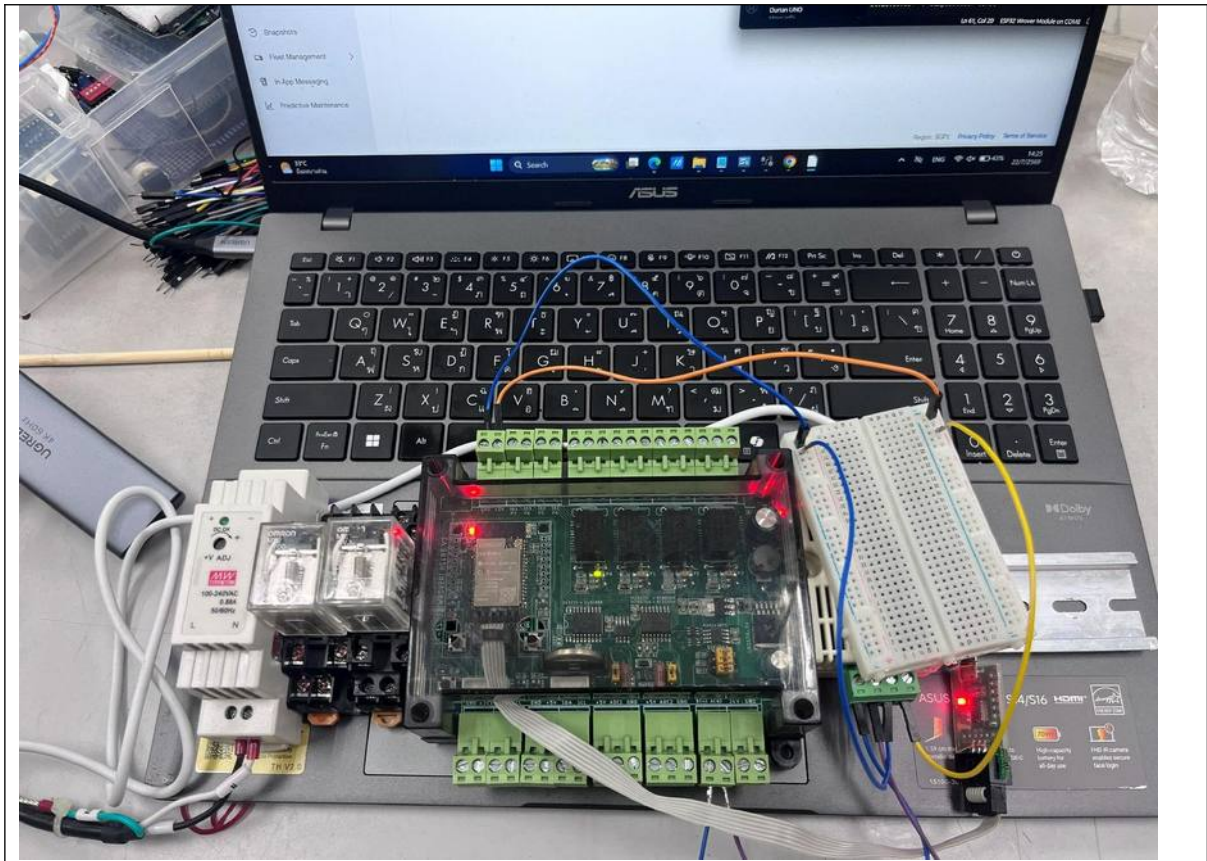
| Transaction  | Address                 | Function | Value |
|--------------|-------------------------|----------|-------|
| Tx:000000-01 | 04 00 01 00 02 20 0B    |          |       |
| Rx:000001-01 | 04 04 01 3B 01 ED 4A 68 |          |       |
| Tx:000002-01 | 04 00 01 00 02 20 0B    |          |       |
| Rx:000003-01 | 04 04 01 3B 01 EC 8B A8 |          |       |
| Tx:000004-01 | 04 00 01 00 02 20 0B    |          |       |
| Rx:000005-01 | 04 04 01 3B 01 EA 0B AA |          |       |
| Tx:000006-01 | 04 00 01 00 02 20 0B    |          |       |
| Rx:000007-01 | 04 04 01 3B 01 E9 4B AB |          |       |
| Tx:000008-01 | 04 00 01 00 02 20 0B    |          |       |
| Rx:000009-01 | 04 04 01 3C 01 E8 3B AA |          |       |
| Tx:000010-01 | 04 00 01 00 02 20 0B    |          |       |
| Rx:000011-01 | 04 04 01 3B 01 E7 CA 6F |          |       |
| Tx:000012-01 | 04 00 01 00 02 20 0B    |          |       |
| Rx:000013-01 | 04 04 01 3B 01 E5 4B AE |          |       |
| Tx:000014-01 | 04 00 01 00 02 20 0B    |          |       |
| Rx:000015-01 | 04 04 01 3F 01 ED 0B A9 |          |       |

## 2. Read >> XY-MD02

- Hardware - B, ESP32 Board

Show with **Blynk2.0**

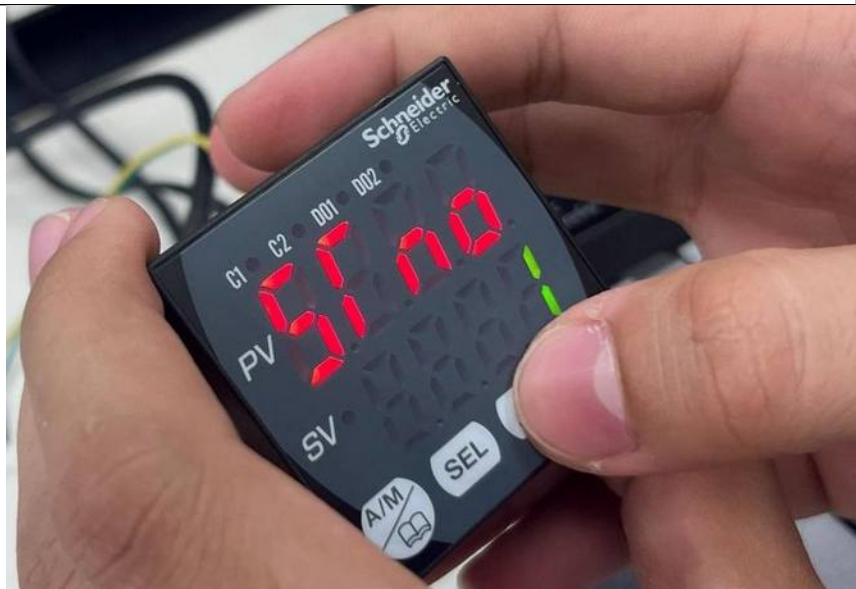
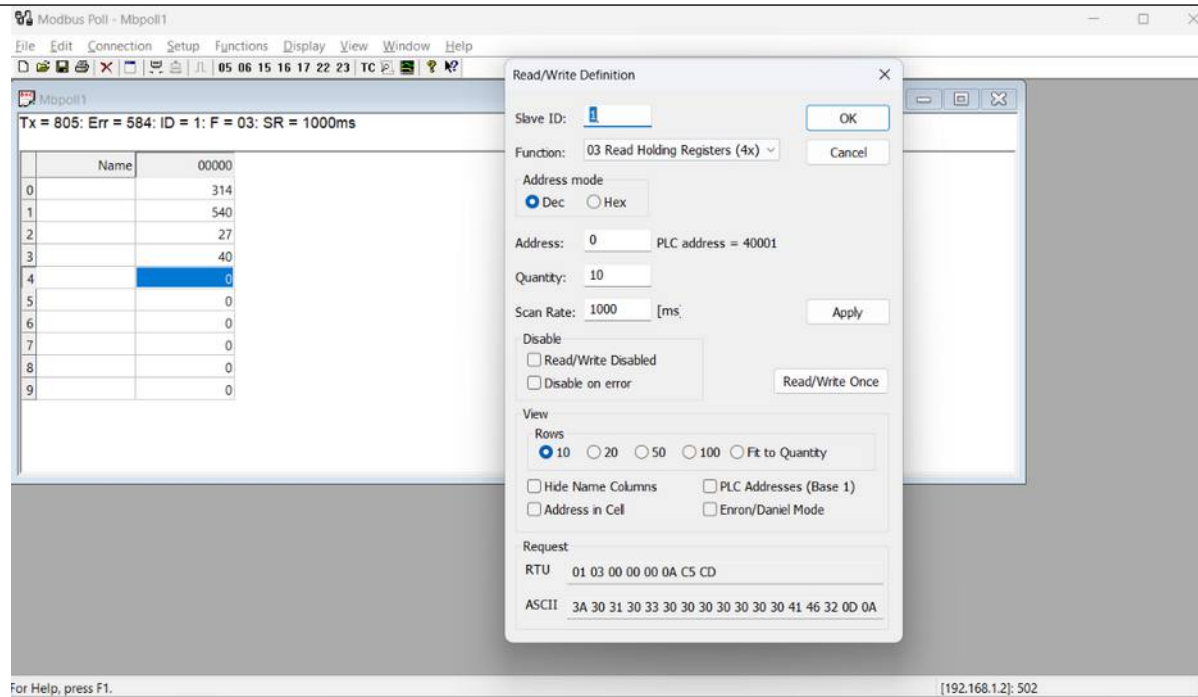




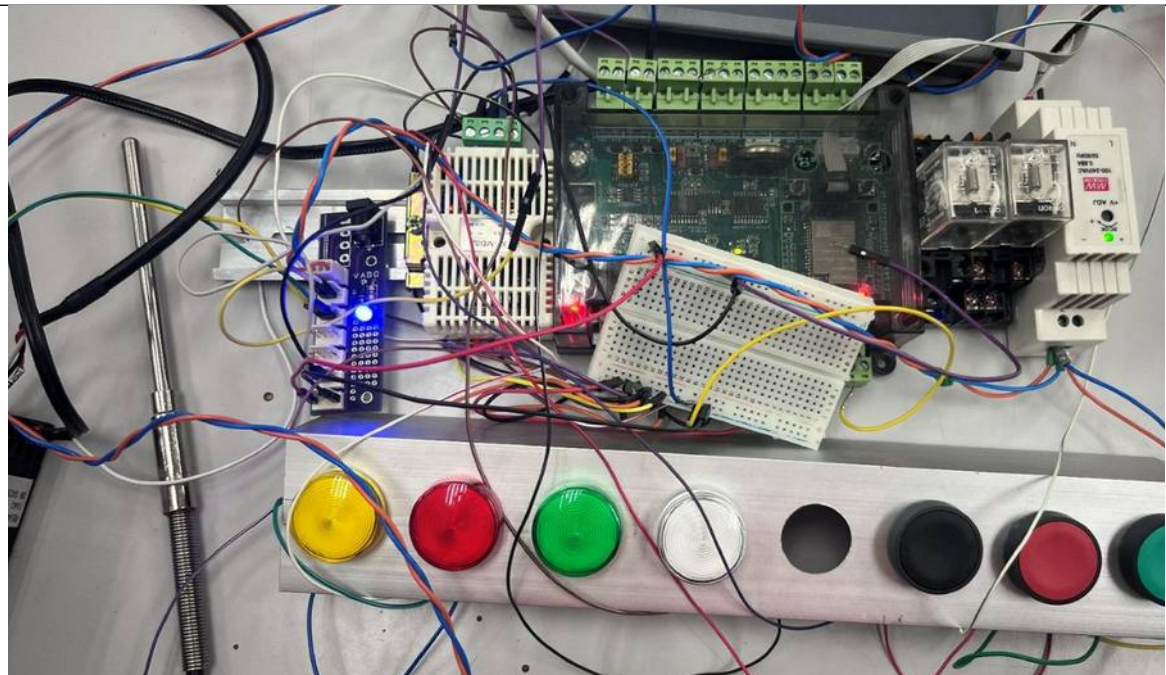
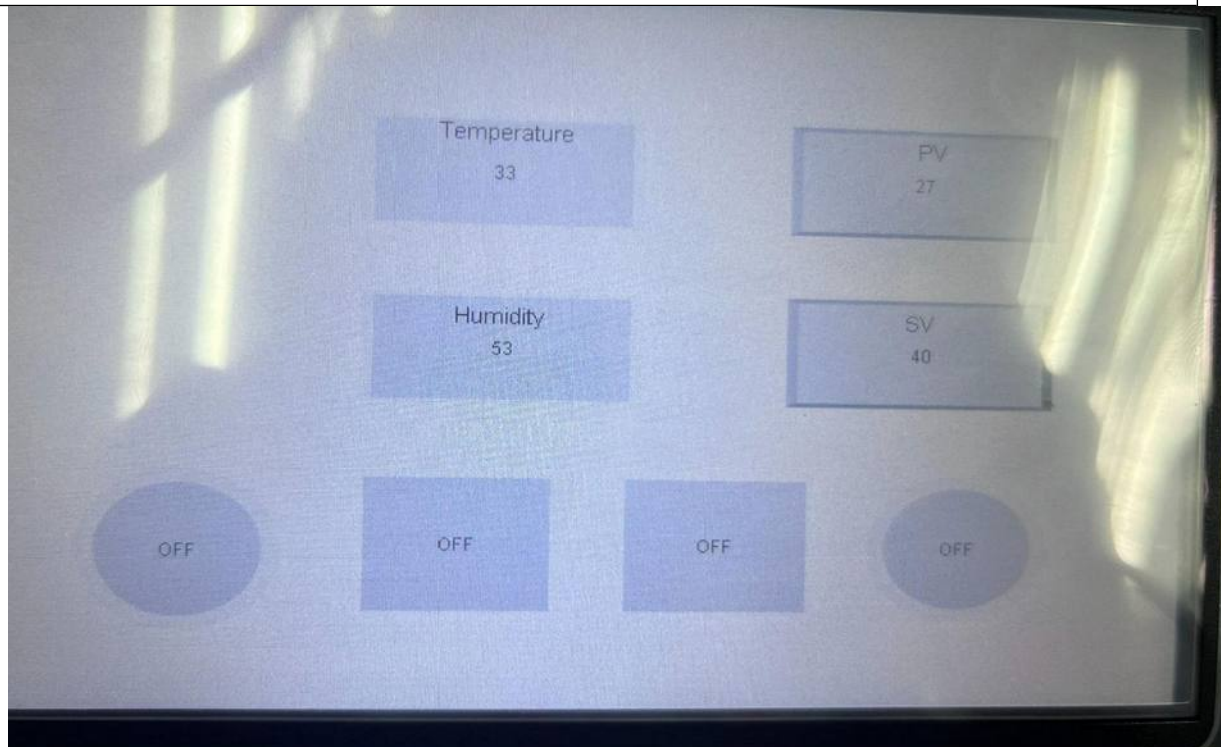
## 7. Modbus TCP

- Hardware - XY
- Show with **Modbus Poll (TCP)**
- Read REG48PUN2LJHU
- Read XY-MD02

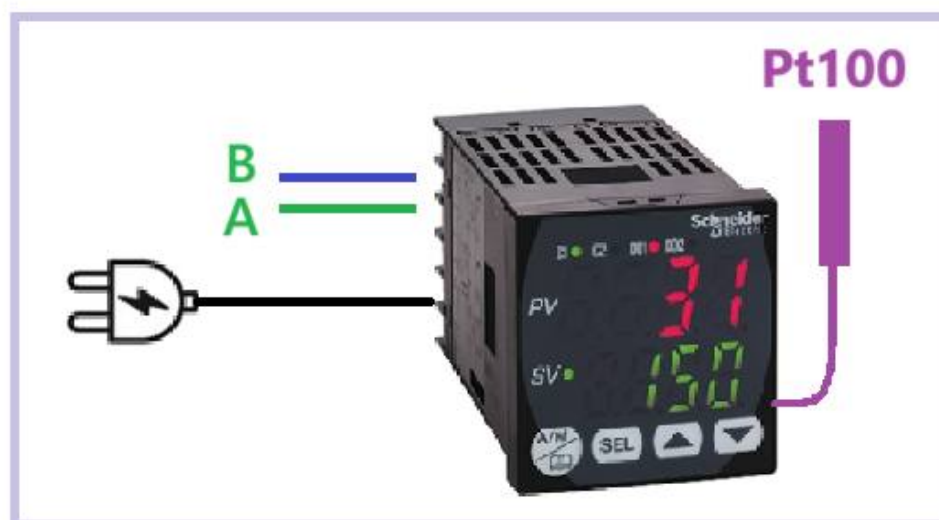
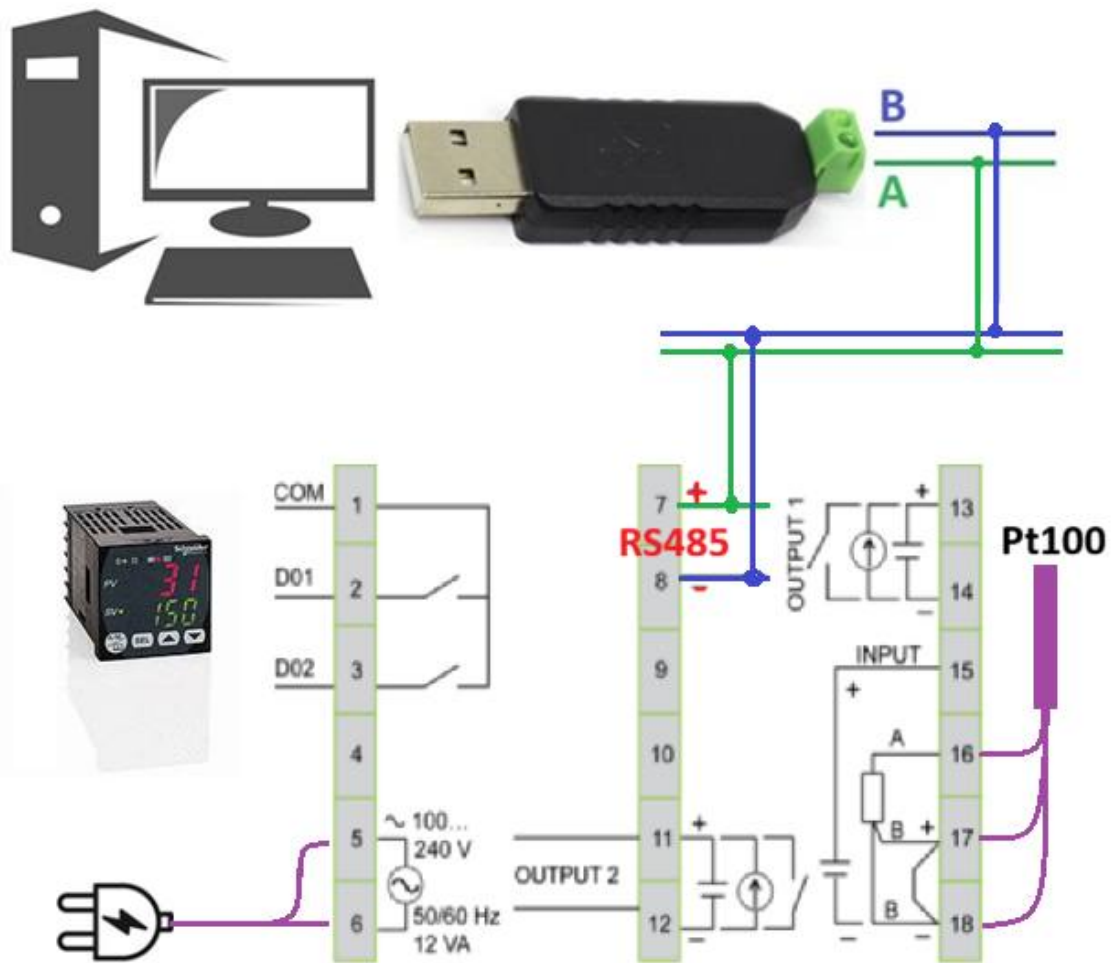
Control Monitor ตามข้อ 10



```
14:20:35.044 -> Temperature: 32.50
14:20:35.076 -> Humidity: 52.40
14:20:35.283 -> PV = 27
14:20:35.325 -> SV = 40
14:20:44.884 -> Temperature: 32.50
14:20:44.884 -> Humidity: 53.20
14:20:45.098 -> PV = 27
14:20:45.134 -> SV = 40
14:20:55.238 -> Temperature: 32.50
14:20:55.238 -> Humidity: 53.40
14:20:55.461 -> PV = 27
14:20:55.461 -> SV = 40
14:21:04.934 -> Temperature: 32.60
14:21:04.934 -> Humidity: 53.30
14:21:05.184 -> PV = 28
14:21:05.184 -> SV = 40
14:21:14.856 -> Temperature: 32.60
```



## Hardware - XS



3. Read >> REG48PUN2LJHU

- Hardware - XS

Show with **Modbus Poll (Serial)**





Modbus Poll - Mbpoll1

File Edit Connection Setup Functions Display View Window Help

05 06 15 16 17 22 23 TC

Mbpoll1

Tx = 805: Err = 584: ID = 1: F = 03: SR = 1000ms

|   | Name | 00000 |
|---|------|-------|
| 0 |      | 314   |
| 1 |      | 540   |
| 2 |      | 27    |
| 3 |      | 40    |
| 4 |      | 0     |
| 5 |      | 0     |
| 6 |      | 0     |
| 7 |      | 0     |
| 8 |      | 0     |
| 9 |      | 0     |

Read/Write Definition

Slave ID:

Function: 03 Read Holding Registers (4x)

Address mode: ☒ Dec ☐ Hex

Address:  PLC address = 40001

Quantity:

Scan Rate:  [ms]

Disable: ☐ Read/Write Disabled ☐ Disable on error

View: Rows ☒ 10 ☐ 20 ☐ 50 ☐ 100 ☐ Fit to Quantity

☐ Hide Name Columns ☐ PLC Addresses (Base 1)

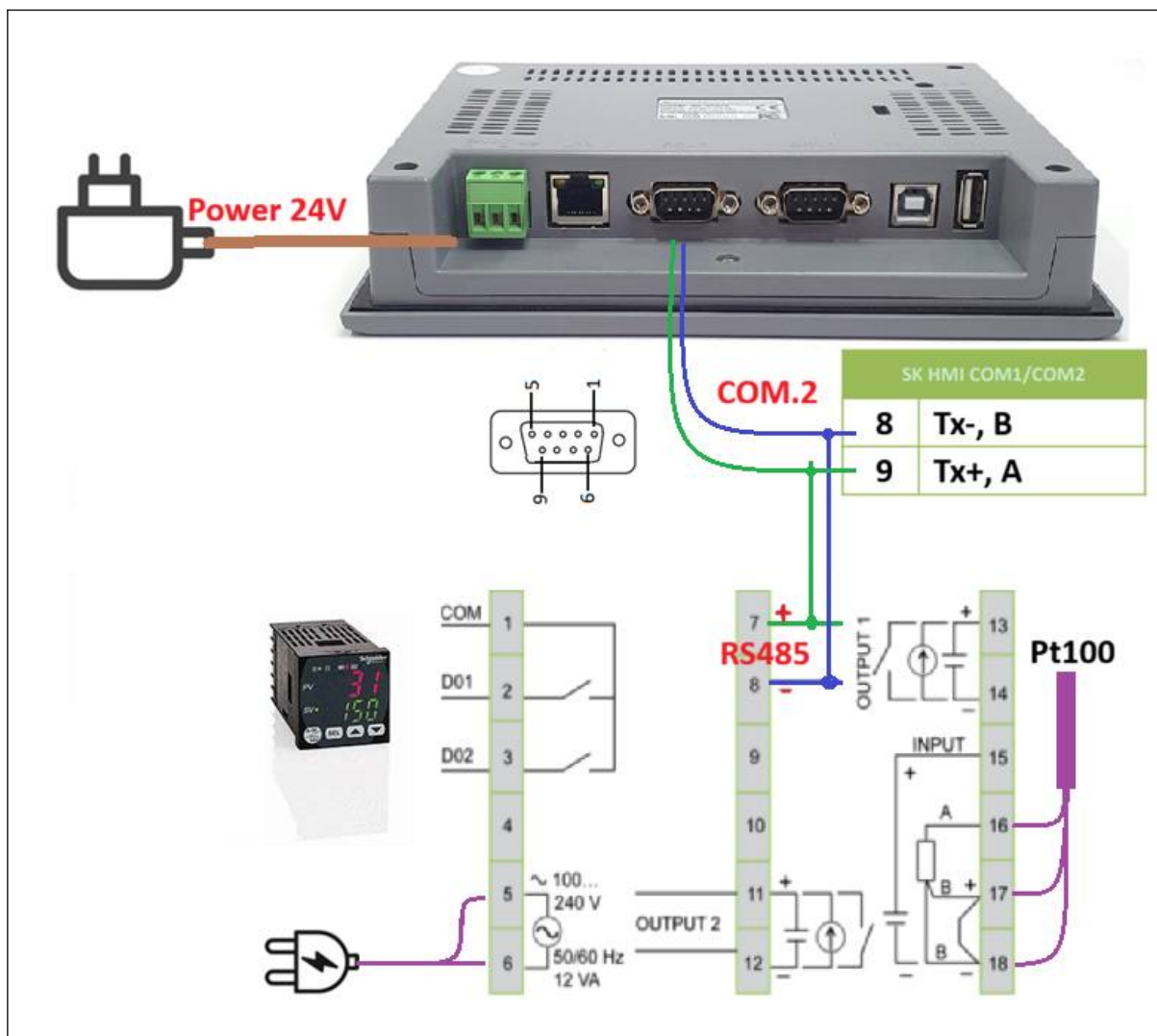
☐ Address in Cell ☐ Error/Daniel Mode

Request: RTU  ASCII

For Help, press F1.

[192.168.1.2]: 502

## Hardware - XU

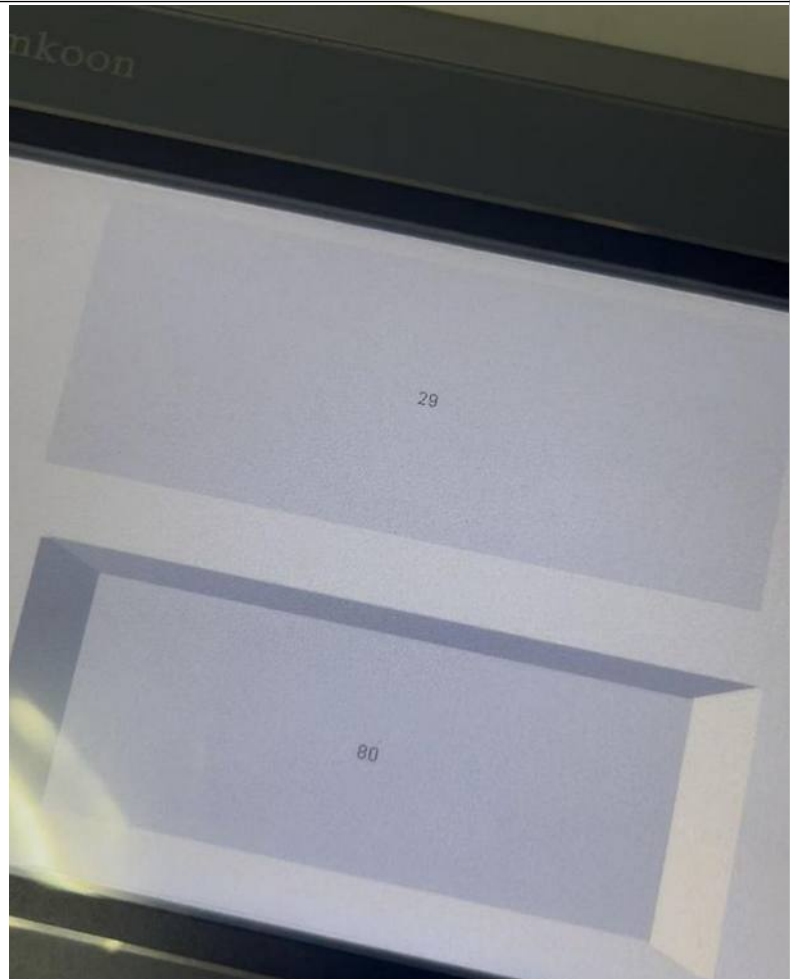


- <https://www.miniature-solution.com/p/127>
- <https://www.samkoon.store/wp-content/uploads/2021/12/SK-User-manual.pdf>

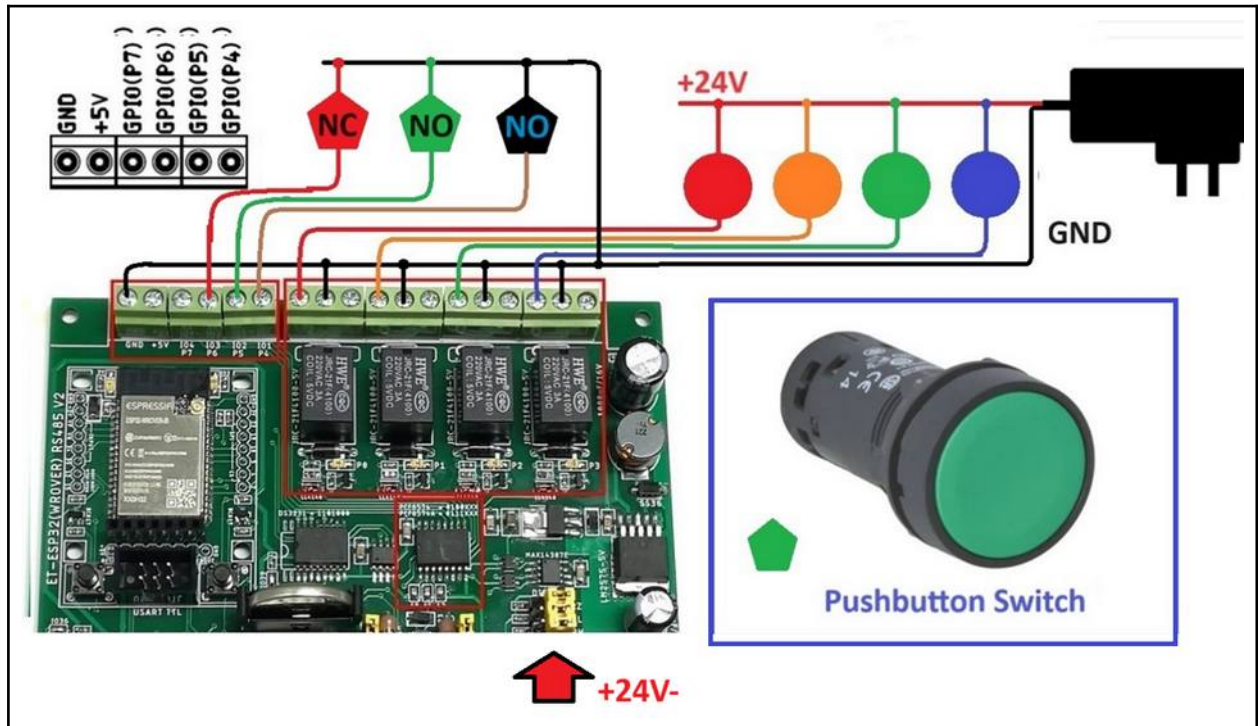
4. Read >> REG48PUN2LJHU

- Hardware - XU

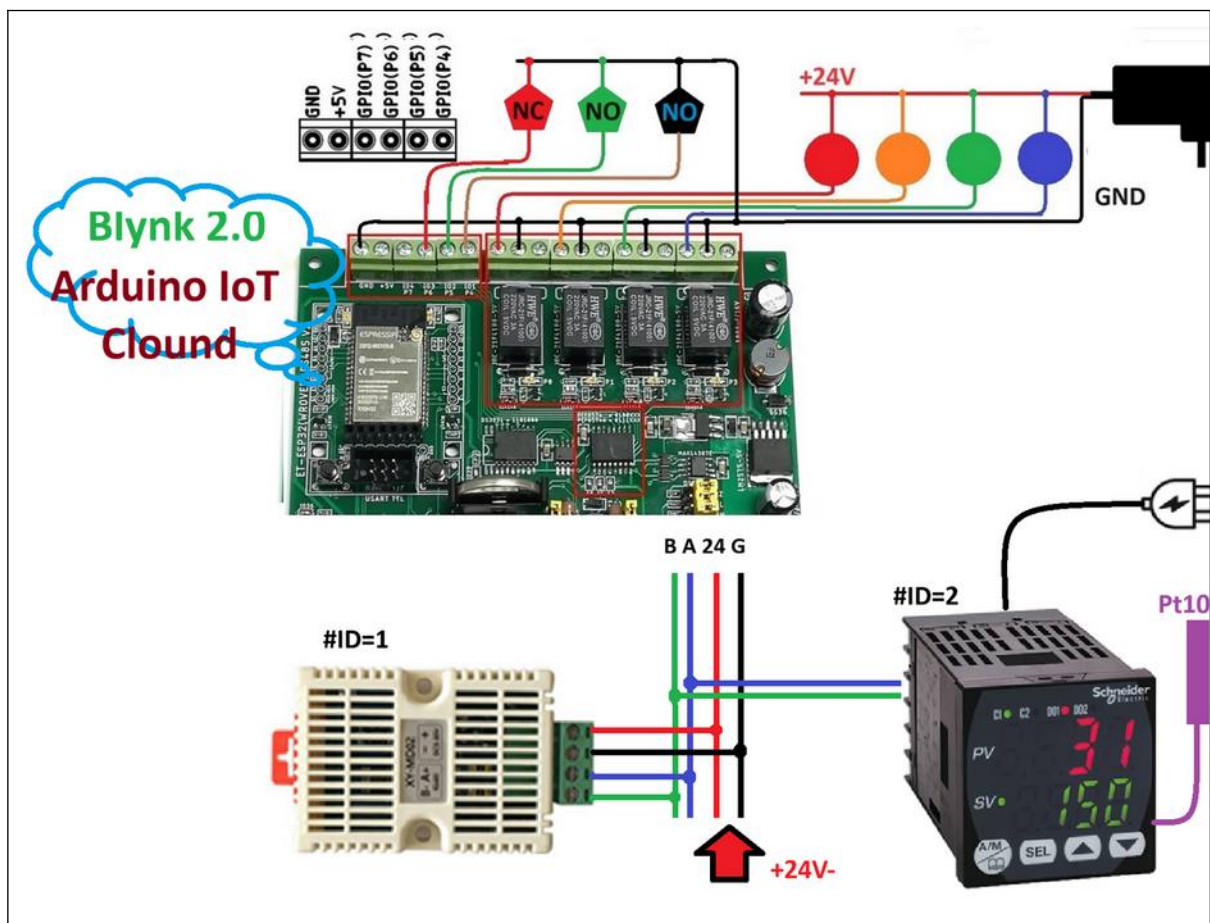
Show with PV, SV on **HMI (SK-070HS) via RS485**



## Hardware - C



## Hardware - XT



#### 5. Control and Monitor

- Hardware - C
- Control Monitor ข้อ 10

Control and monitor by **blynk**

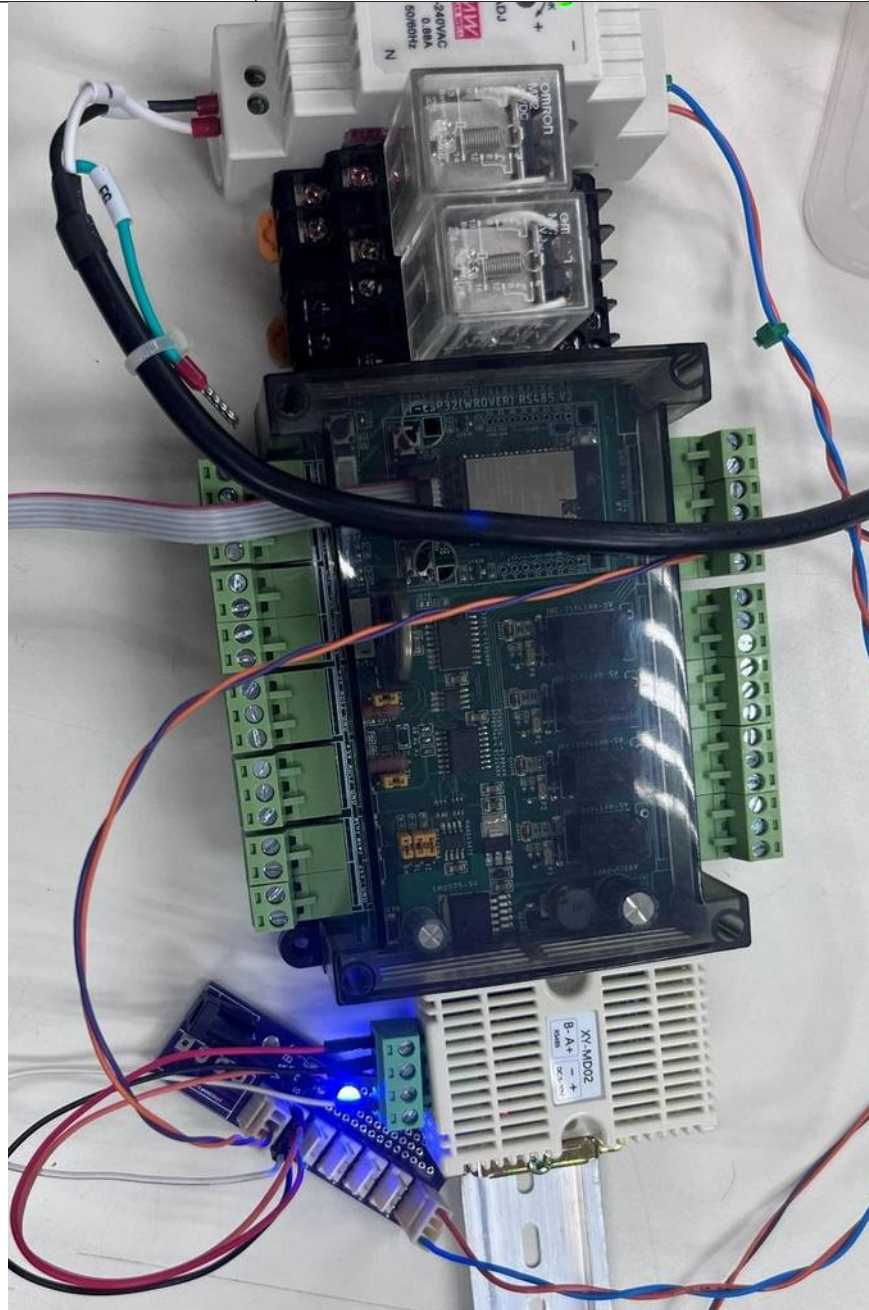
- **Arduino IoT Platform**

#### -6. Control and Monitor

- Hardware - XT
- Control Monitor ข้อ 10
- Display REG48PUN2LJHU
- Display XY-MD02

Control and monitor by **blynk**

**Arduino IoT Platform**



Devices - Blynk.Console    Modbus Address Error    modbus poll xy md02 - &url=    +

blynk.cloud/dashboard/1018842/global/devices/227686/organization/1018842/devices/4139017/dashboard

**Blynk.Console**

My organization - 2945WT

**EmbeddedW05 5and6** Online  
Auth Token: \*\*\*\*\*-31ZF    Ma    My organization - 2945WT

Live 1h 6h 1d 1w 1mo 3mo 6mo 1y

Write Lamp ☐    Write Lamp ☐    Leay ☐    Red Lamp ☒

Green Lamp ☒    Green Lamp ☒    Relay2 ☐    Yellow Lamp ☒

Humidity [XY-MD02]    Temp [XY-MD02]

56.4%    30.4°C

Output Serial Monitor X

Message (Enter to send message to ESP32 Wrover Module on /COM5)

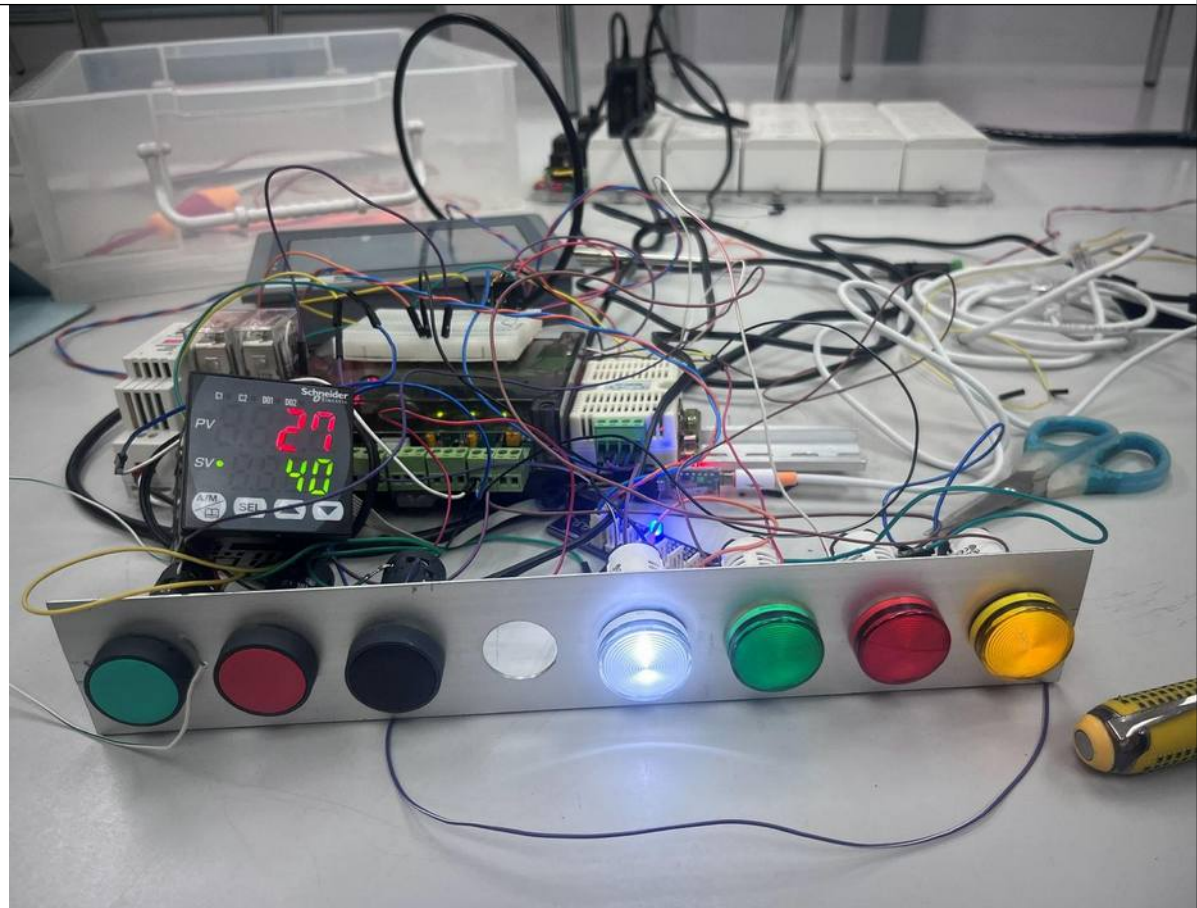
```
15:57:08.028 -> Temperature: 30.60
15:57:08.060 -> Humidity: 57.00
15:57:08.348 -> PV = 27
15:57:08.348 -> SV = 40
15:57:18.029 -> Temperature: 30.10
15:57:18.061 -> Humidity: 56.40
15:57:18.333 -> PV = 27
15:57:18.367 -> SV = 40
15:57:28.058 -> Temperature: 30.20
15:57:28.058 -> Humidity: 56.70
15:57:28.366 -> PV = 27
15:57:28.366 -> SV = 40
15:57:38.032 -> Temperature: 30.30
15:57:38.063 -> Humidity: 56.40
15:57:38.360 -> PV = 27
15:57:38.360 -> SV = 40
15:57:48.064 -> Temperature: 30.40
15:57:48.064 -> Humidity: 56.20
15:57:48.360 -> PV = 27
15:57:48.360 -> SV = 40
```

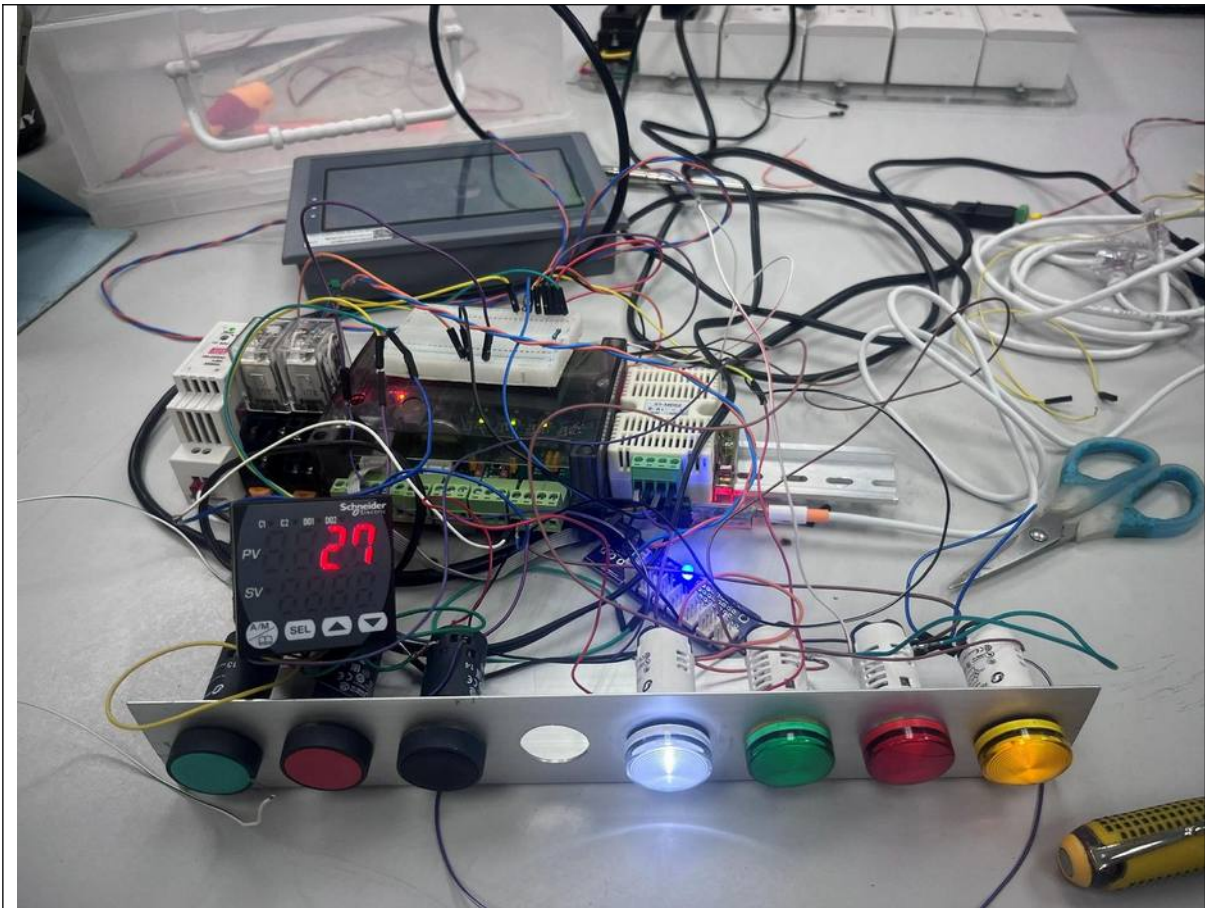
Ln 132, Col 44    ESP32 Wrover Module

27  
SV  
40

Region: SGP1    Privacy Policy    Terms of Service

32°C Easton, NJ    Search    ENG    100%    15:57 2/8/2569





Modbus Poll - Mbpoll1

File Edit Connection Setup Functions Display View Window Help

05 06 15 16 17 22 23 TC

Mbpoll1

Tx = 3: Err = 0: ID = 2: F = 04: SR = 1000ms

|   | Name | 01000 |
|---|------|-------|
| 0 |      | 27    |
| 1 |      | 40    |
| 2 |      |       |
| 3 |      |       |
| 4 |      |       |
| 5 |      |       |
| 6 |      |       |
| 7 |      |       |
| 8 |      |       |
| 9 |      |       |

Read/Write Definition

Slave ID:

Function: 04 Read Input Registers (3x)

Address mode  
☒ Dec ☐ Hex

Address:  PLC address = 31001

Quantity:

Scan Rate:  [ms]

Disable  
☐ Read/Write Disabled  
☐ Disable on error

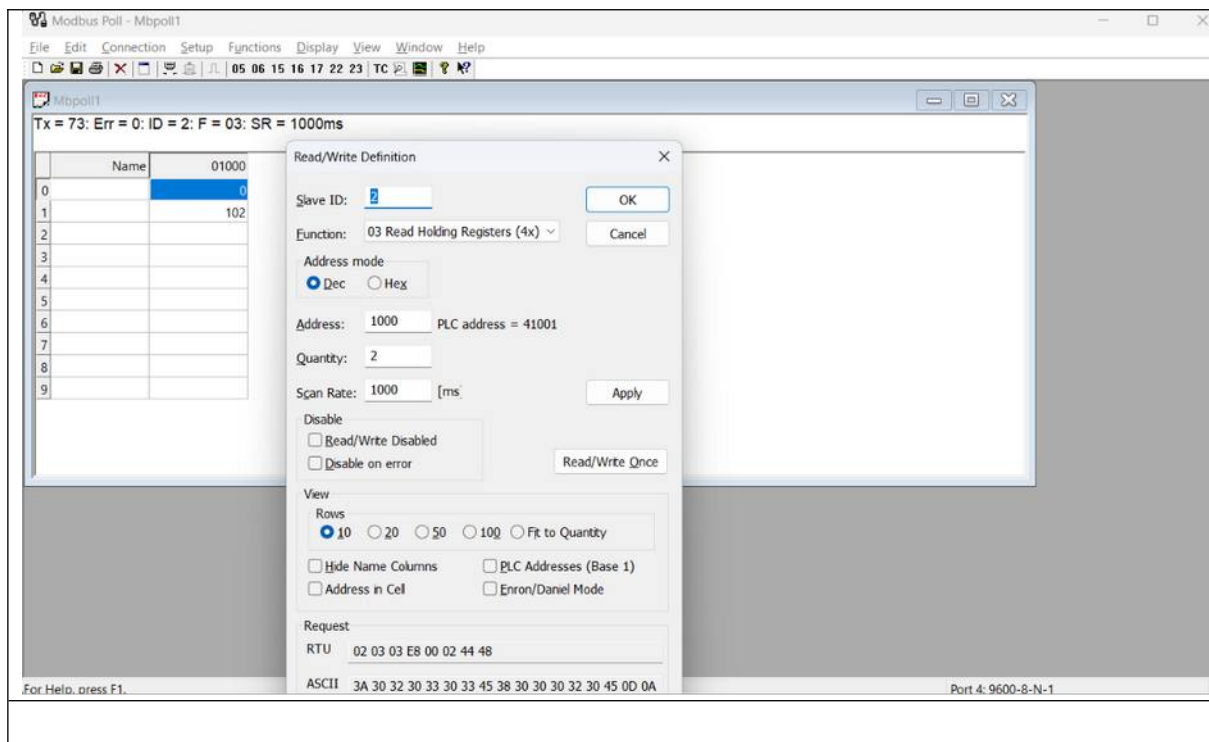
View  
Rows  
☒ 10 ☐ 20 ☐ 50 ☐ 100 ☐ Fit to Quantity  
☐ Hide Name Columns ☐ PLC Addresses (Base 1)  
☐ Address in Cell ☐ Enron/Daniel Mode

Request  
RTU 02 04 03 E8 00 02 F1 88  
ASCII 3A 30 32 30 34 30 33 45 38 30 30 32 30 44 0D 0A

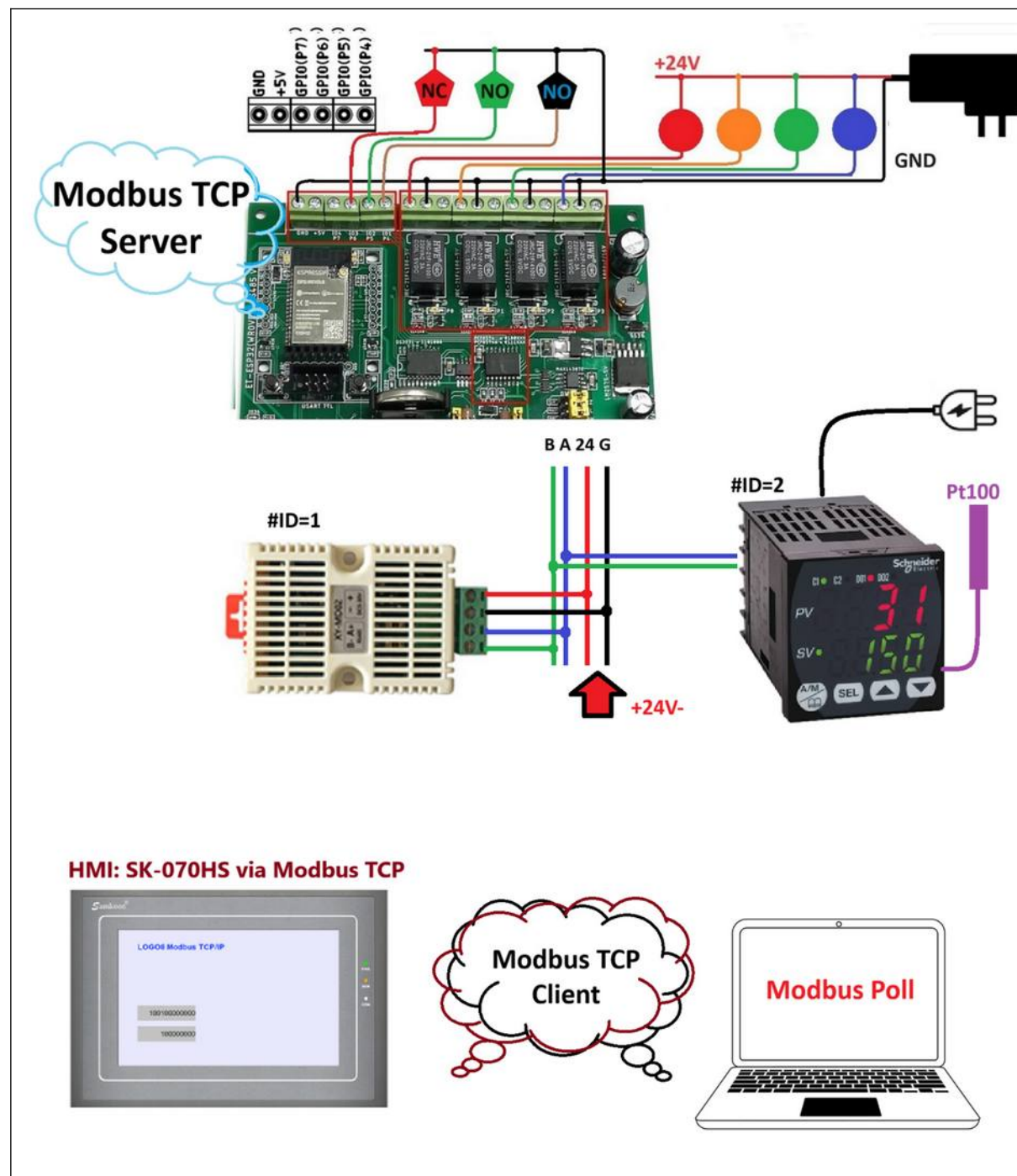
OK Cancel Apply Read/Write Once

For Help, press F1.

Port 4: 9600-R-N-1



## Hardware - XZ

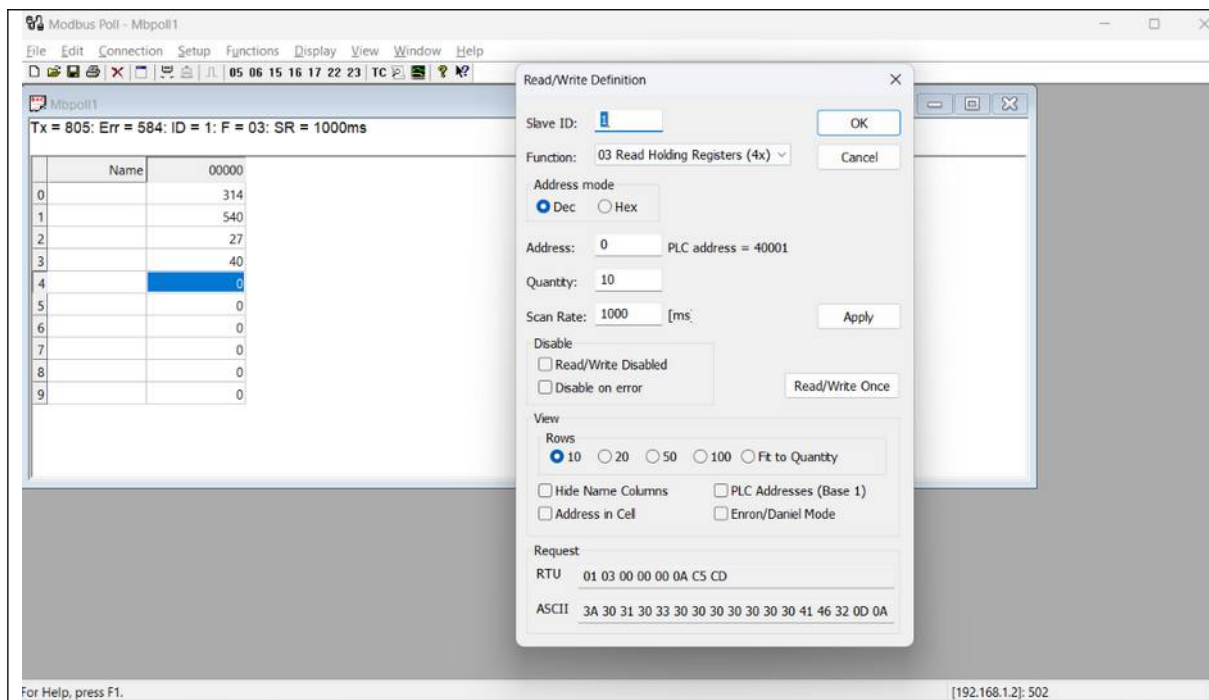


#### 8. Modbus TCP with HMI

- Hardware - XZ
- Show with **HMI(SK-070HS) via Modbus TCP**
- Read REG48PUN2LJHU
- Read XY-MD02

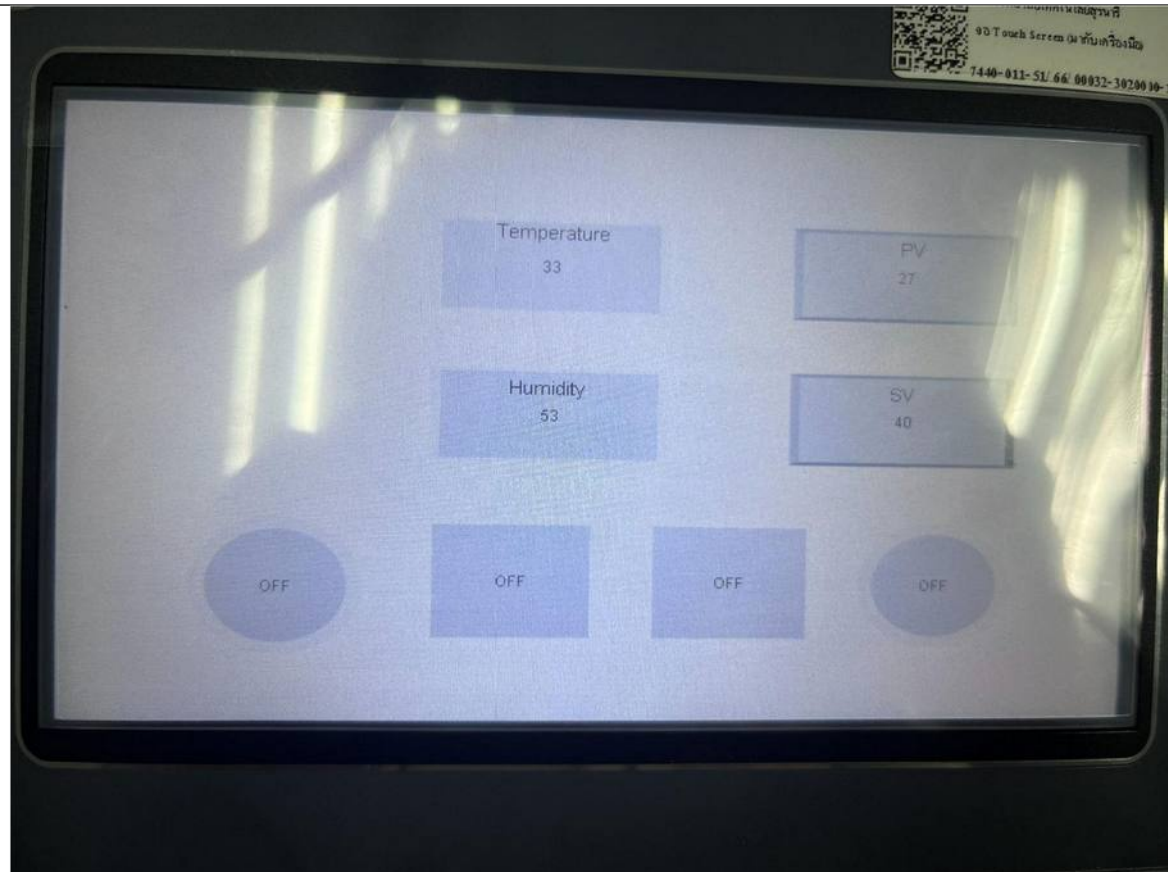
Control Monitor ตามข้อ 10





For Help, press F1.

[192.168.1.2]: 502



```
14:20:35.044 -> Temperature: 32.50
14:20:35.076 -> Humidity: 52.40
14:20:35.203 -> PV = 27
14:20:35.325 -> SV = 40
14:20:44.884 -> Temperature: 32.50
14:20:44.884 -> Humidity: 53.20
14:20:45.098 -> PV = 27
14:20:45.134 -> SV = 40
14:20:55.238 -> Temperature: 32.50
14:20:55.238 -> Humidity: 53.40
14:20:55.461 -> PV = 27
14:20:55.461 -> SV = 40
14:21:04.934 -> Temperature: 32.60
14:21:04.934 -> Humidity: 53.30
14:21:05.104 -> PV = 28
14:21:05.104 -> SV = 40
14:21:14.856 -> Temperature: 32.60
14:21:14.898 -> Humidity: 52.60
14:21:15.120 -> PV = 28
14:21:15.157 -> SV = 40
14:21:25.128 -> Temperature: 32.60
14:21:25.163 -> Humidity: 52.10
14:21:25.322 -> PV = 28
14:21:25.359 -> SV = 40
14:21:34.861 -> Temperature: 32.60
14:21:34.913 -> Humidity: 51.90
14:21:35.114 -> PV = 28
14:21:35.114 -> SV = 40
14:21:45.306 -> Temperature: 32.60
14:21:45.340 -> Humidity: 51.80
14:21:45.532 -> PV = 28
14:21:45.532 -> SV = 40
```